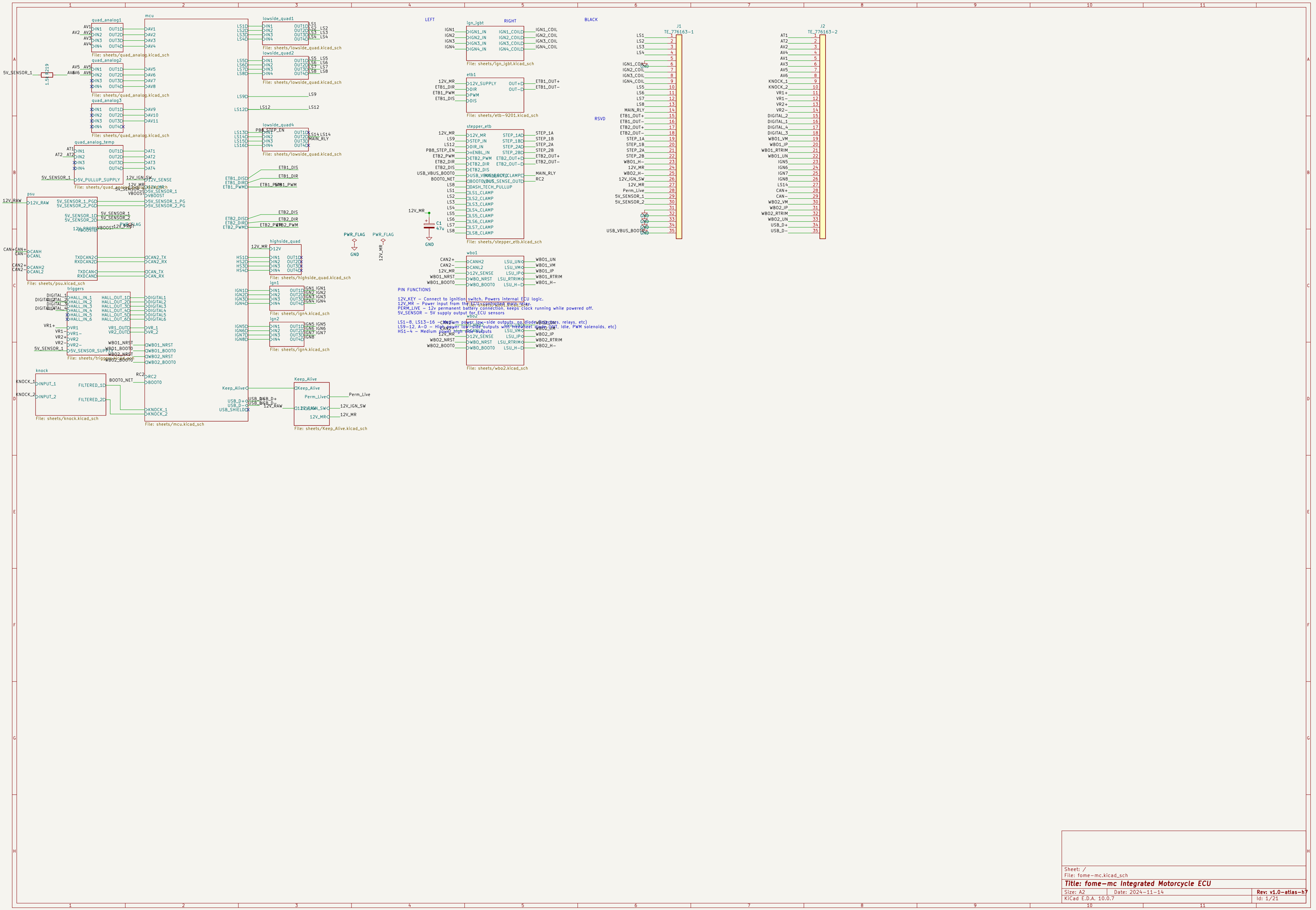
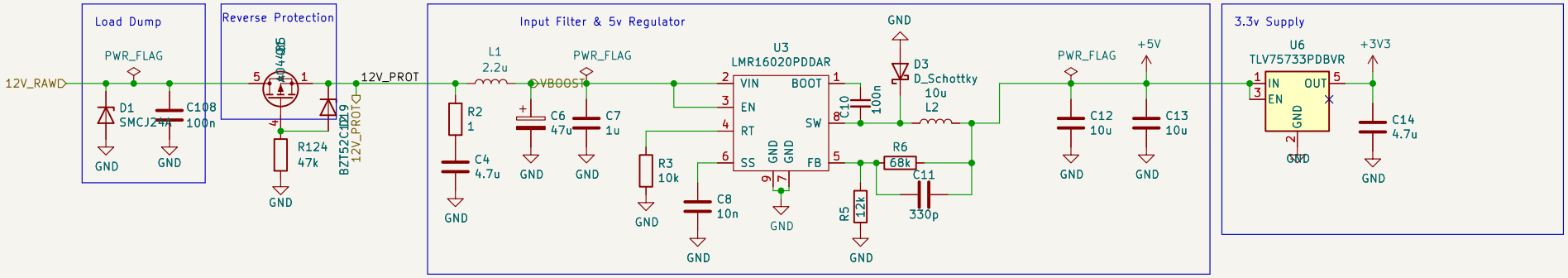
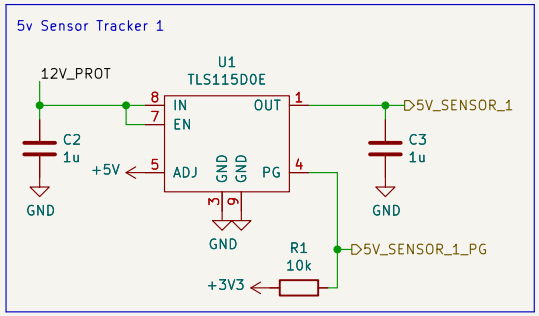


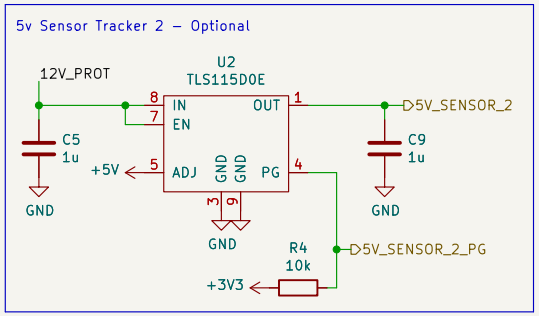
Sheet: /
File: fome-mc.kicad_sch

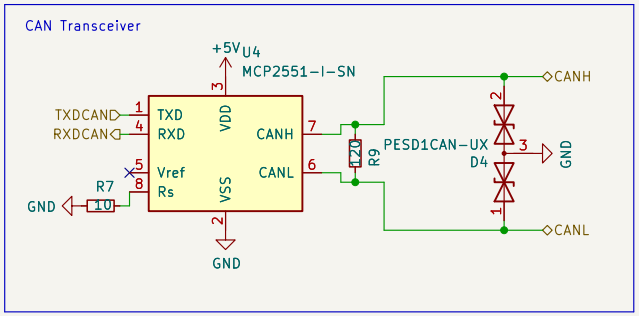
Title: fome-mc Integrated Motorcycle ECU

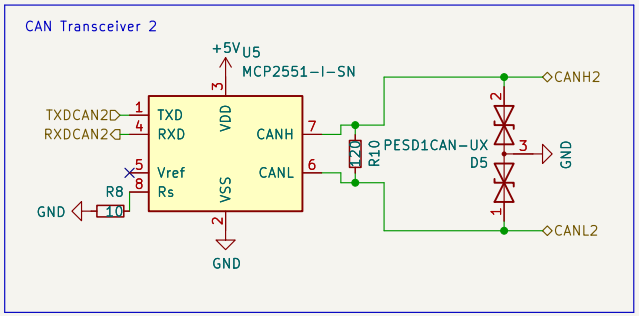
Size: A2	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 1/21









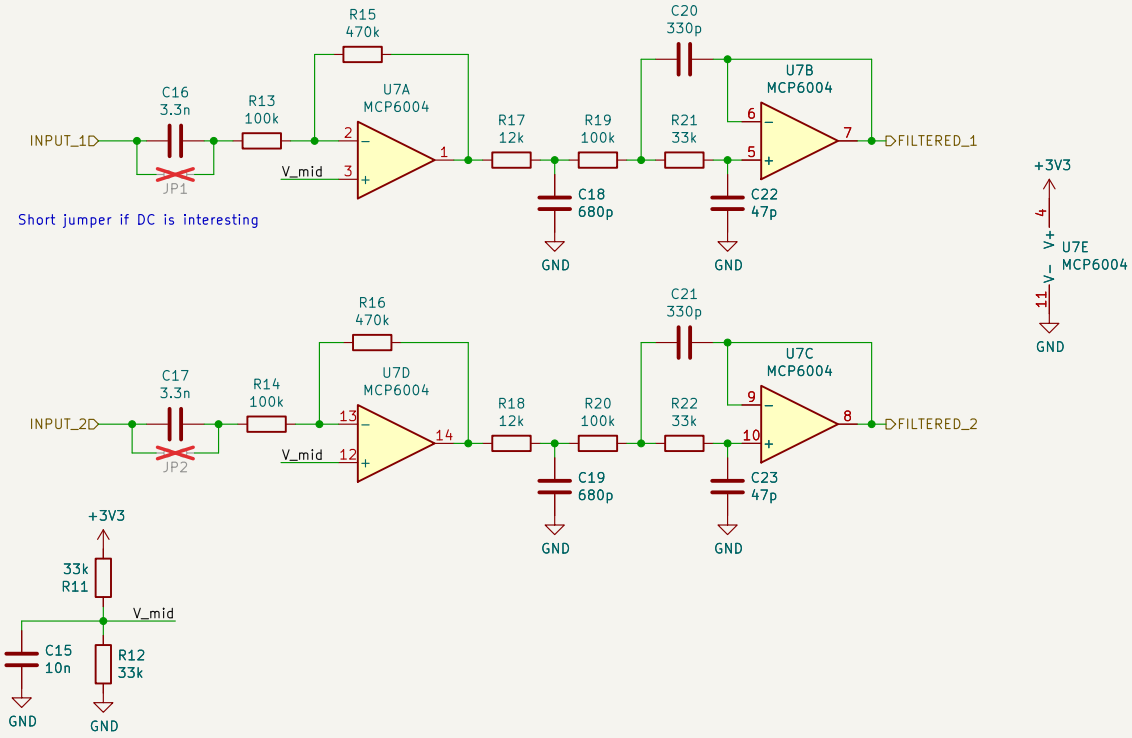


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Sheet: /psu/
File: psu.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 2/21

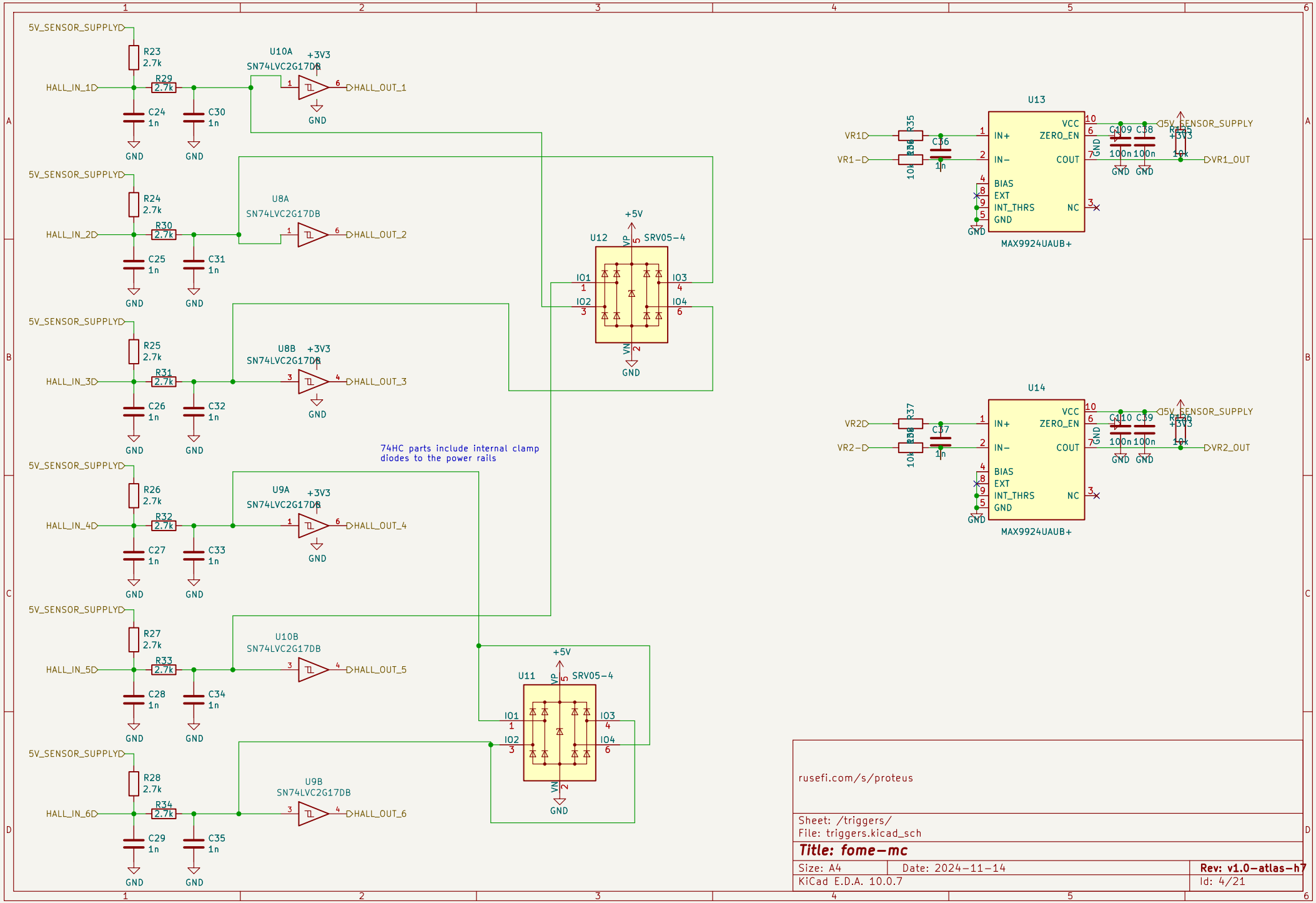


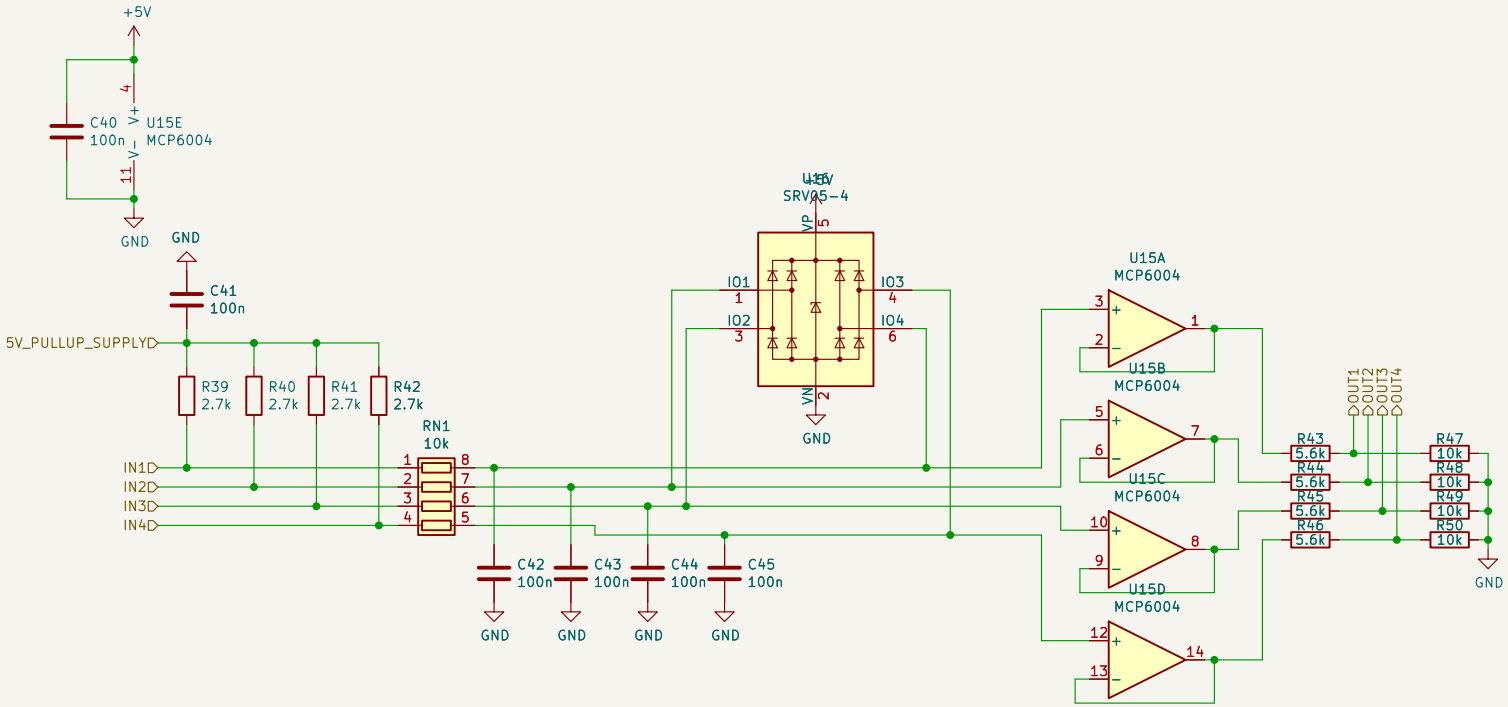
rusefi.com/s/proteus

Sheet: /knock/
File: knock.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 3/21


5V_SENSOR_SUPPLY
HALL_IN_1
HALL_IN_2
HALL_IN_3
HALL_IN_4
HALL_IN_5
HALL_IN_6
R23 2.7k
R29 2.7k
R24 2.7k
R30 2.7k
R25 2.7k
R31 2.7k
R26 2.7k
R32 2.7k
R27 2.7k
R33 2.7k
R28 2.7k
R34 2.7k
C24 1n
C30 1n
C25 1n
C31 1n
C26 1n
C32 1n
C27 1n
C33 1n
C28 1n
C34 1n
C29 1n
C35 1n
GND
U10A
SN74LVC2G17DB
+3V3
HALL_OUT_1
U8A
HALL_OUT_2
U8B
HALL_OUT_3
U9A
HALL_OUT_4
U10B
HALL_OUT_5
U9B
HALL_OUT_6
74HC parts include internal clamp diodes to the power rails
U12
SRV05-4
+5V
IO1
IO2
IO3
IO4
VP
VN
U11
U13
MAX9924UAUB+
VR1
VR1-
R35
R36
10k
C36
1n
IN+
IN-
VCC
ZERO_EN
COUT
BIAS
EXT
INT_THRS
NC
C109
C38
100n
VR1_OUT
U14
MAX9924UAUB+
VR2
VR2-
R37
R38
C37
C110
C39
VR2_OUT
rusefi.com/s/proteus
Sheet: /triggers/
File: triggers.kicad_sch
Title: fome-mc
Size: A4
Date: 2024-11-14
KiCad E.D.A. 10.0.7
Rev: v1.0-atlas-h7
Id: 4/21

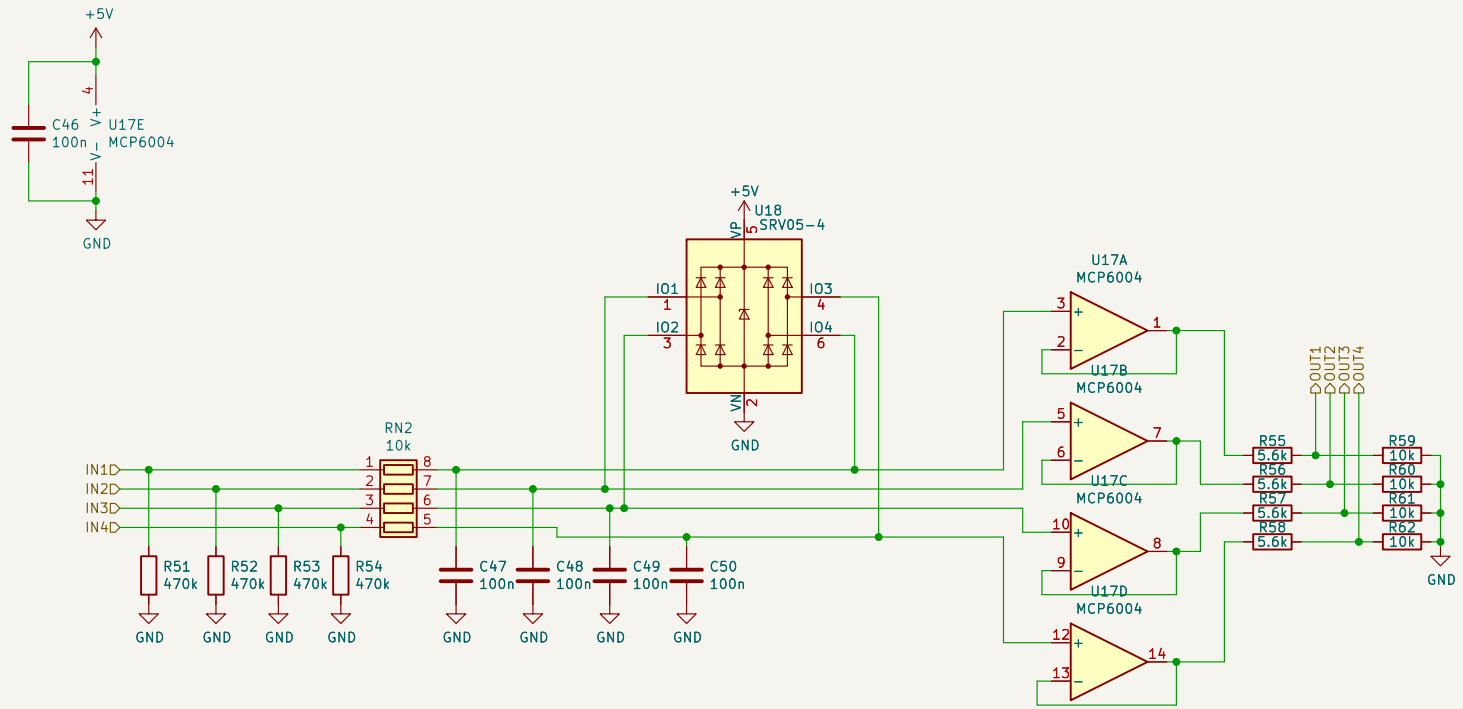


rusefi.com/s/proteus

Sheet: /quad_analog_temp/
File: quad_analog_pullup.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 5/21

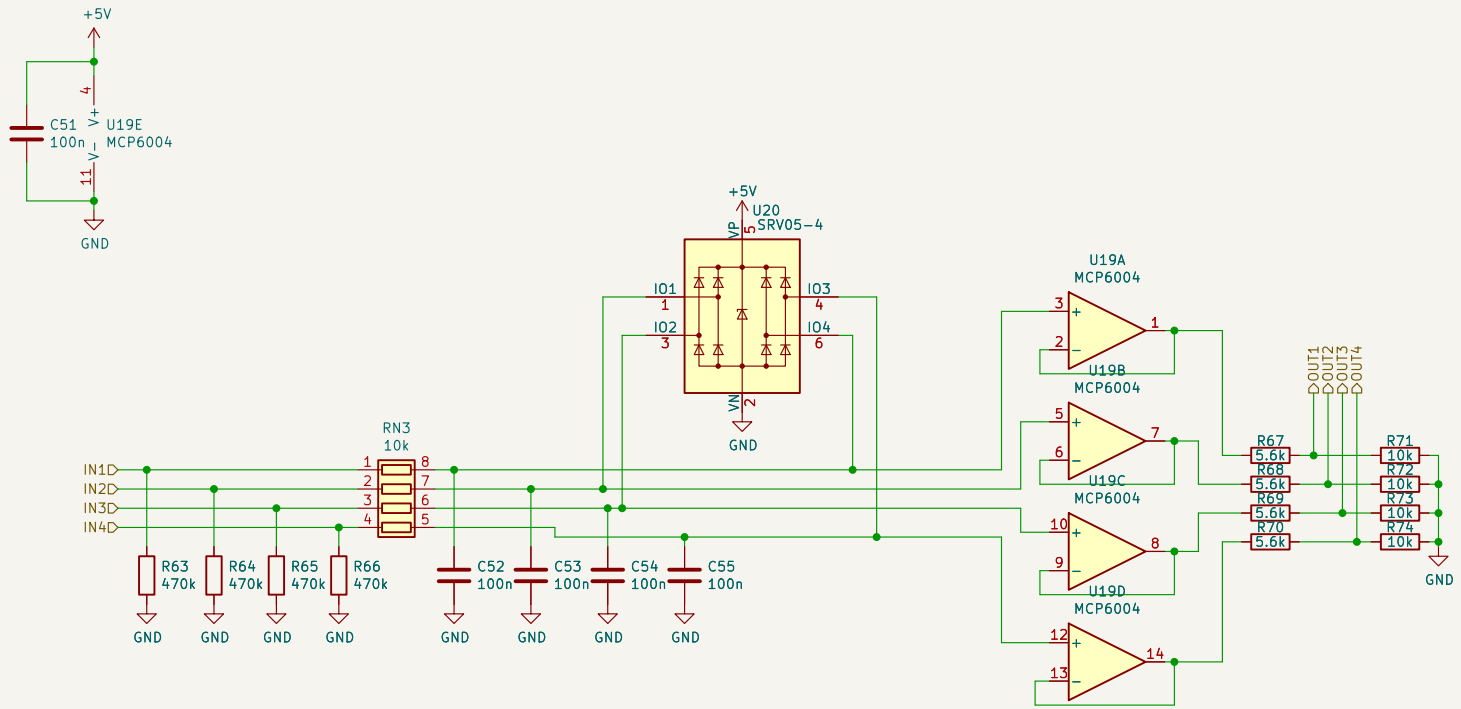


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Sheet: /quad_analog1/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 6/21

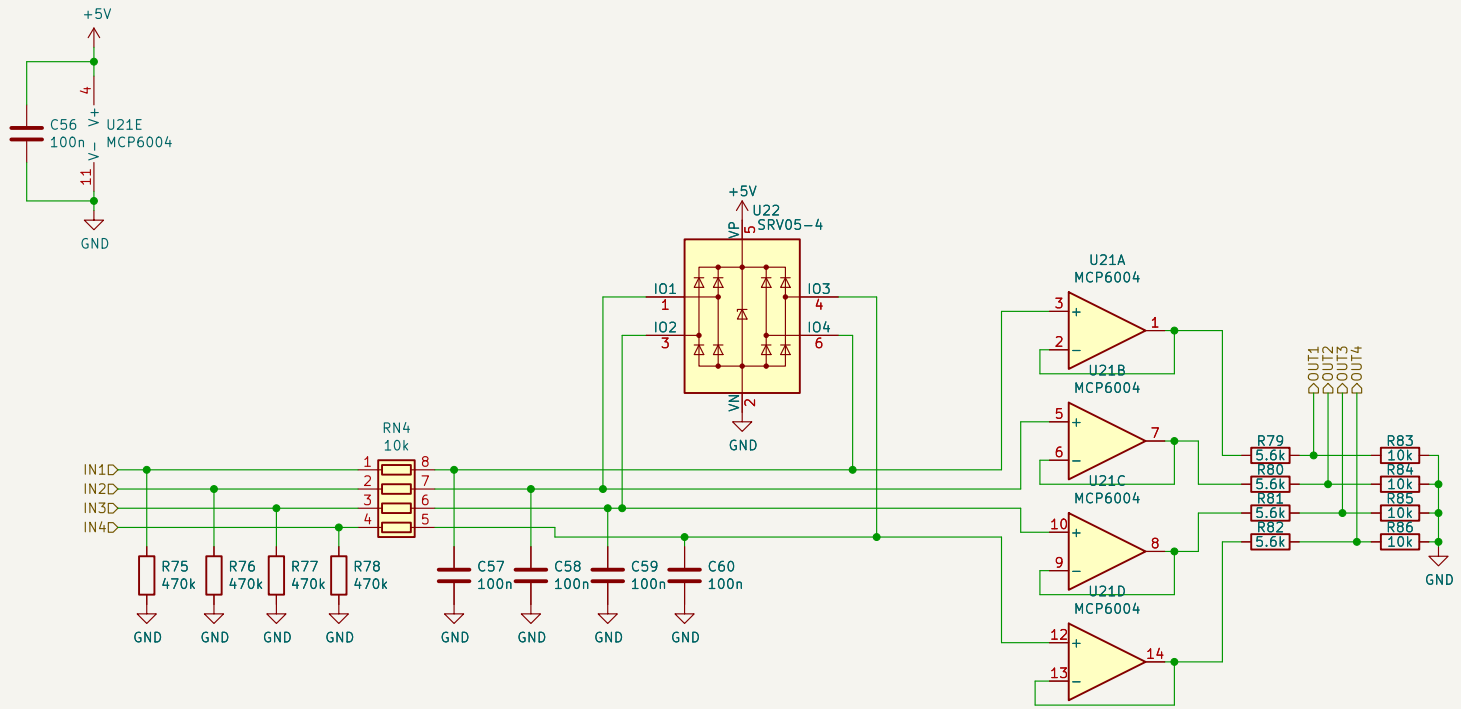


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Sheet: /quad_analog2/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 7/21



rusefi.com/s/proteus

Sheet: /quad_analog3/
File: quad_analog.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 8/21

C61 100n +3V3
C62 100n +3V3
C63 100n +3V3
C64 100n +3V3
C65 4.7u +3V3
C66 4.7u +3V3
C67 100n +3V3
C68 100n +3V3
C69 100n +3V3
C71 100n +3V3
C72 4.7u +3V3
C73 100n +3V3
C76 4.7u +3V3

BT1 Battery_Cell
B5819W D6
Keep_Alive
D7 B5819W
C80 4.7u
D8 B5819W
PWR_FLAG
VDDA
FB1 Ferrite_Bead
+3V3
C81 100n GND
C82 4.7u GND

SW1 GND
C79 100n nRESET GND
R89 10k +3V3

U23 REF3333 +5V GND OUT
C70 100n GND
C74 33p GND
C75 33p GND
Y1 8MHz

+3V3 BOOT0 138
BOOT0 32
VREF+ 23
PH0 24
PH1

NRST 25
PDR_ON 143

VBAT 6
VDD 17 30 39 52 62 72 84 108 131 144 33
VDDA 121
VDDSDMMC 95
VDDUSB

STM32H753ZIT6
U26

PA0 34 AV1
PA1 35 AV2
PA2 36 AV3
PA3 37 AV4
PA4 40 AV5
PA5 41 AV6
PA6 42 AV7
PA7 43 AV8
PA8 100
PA9 101 RC2
PA10 102
PA11 103 USB_D−
PA12 104 USB_D+
PA13 105 SWDIO
PA14 109 SWCLK
PA15 110 LS7

PB0 46 AV9
PB1 47 AV10
PB2 48 GND
PB3 133 IGN1
PB4 134 LS9
PB5 135
PB6 136
PB7 137 LS12
PB8 139 LS13
PB9 140 LS14
PB10 69 BARO_SCL
PB11 70 BARO_SDA
PB12 73 CAN2_RX
PB13 74 CAN2_TX
PB14 75
PB15 76

PC0 26 AV11
PC1 27 AT1
PC2 28 AT2
PC3 29 AT3
PC4 44 AT4
PC5 45 12V_DIVIDED
PC6 96 LS5
PC7 97 LS6
PC8 98 SD_D0
PC9 99 SD_D1
PC10 111 SD_D2
PC11 112 SD_D3
PC12 113 SD_CLK
PC13 7 WIFI_RST
PC14 8
PC15 9

PD0 114 CAN_RX
PD1 115 CAN_TX
PD2 116 SD_CMD
PD3 117 LS8
PD4 118 WBO1_NRST
PD5 119 WBO1_BOOT0
PD6 122 WBO2_NRST
PD7 123 WBO2_BOOT0
PD8 77 ETB2_DIS
PD9 78 ETB2_DIR
PD10 79 ETB1_DIR
PD11 80 ETB1_DIS
PD12 81 ETB1_PWM
PD13 82 ETB2_PWM
PD14 85 LED3
PD15 86 HS4

PF0 10 LED1
PF1 11 LED4
PF2 12 LED2
PF3 13
PF4 14 KNOCK_1D
PF5 15 KNOCK_2D
PF6 18 AUX_SPI_CS
PF7 19 AUX_SPI_SCK
PF8 20 AUX_SPI_MISO
PF9 21 AUX_SPI_MOSI
PF10 22
PF11 49 MR_DIVIDED
PF12 50 5V_SENSE_DIVIDED
PF13 53 VBOOST_DIVIDED
PF14 54 5V_SENSOR_2_PGD
PF15 55 5V_SENSOR_1_PGD

PE0 141 LS15
PE1 142 LS16
PE2 1 WIFI_SCK
PE3 2 WIFI_IRQ
PE4 3 WIFI_CS
PE5 4 WIFI_MISO
PE6 5 WIFI_MOSI
PE7 58
PE8 59
PE9 60
PE10 63 DIGITAL3D
PE11 64 DIGITAL4D
PE12 65 DIGITAL5D
PE13 66 DIGITAL6D
PE14 67 VR_1D
PE15 68 VR_2D

VCAP_1 71
VCAP_2 106
C77 2.2u GND
C78 2.2u GND

VSS 16 31
VSSA

Cortex Debug Header
J8 Conn_02x05_Odd_Even
+3V3 SWDIO SWCLK nRESET GND

Tag Connect Debug Header
J9 Conn_01x06
+3V3 SWDIO nRESET SWCLK

12V Sense Divider
12V_SENSED
R116 82k 12V_SENSED
MR_DIVIDED R117 10k C103 100n GND
R118 10k 5V_SENSE_DIVIDED R119 10k C104 100n GND
R120 82k VBOOST_DIVIDED R121 10k C105 100n GND
R96 82k
R97 10k GND
C84 1u GND
12V_DIVIDED IO1 IO2 IO3 IO4
D15 SRV05-4 +3V3 VP VN GND

USB
D14 B5819W +5V
USB_SHIELD
C83 100n GND
R98 100k GND
USBLC6-2SC6 U27 USB_D+ USB_D− GND
Pin header
USB_D− USB_D+ GND J13 USB_B
Connector side
USB_D− USB_D+ GND J14 USB_B

SW2
J15 Conn_01x01
J16 Conn_01x01
BOOT0
R90 100k GND

D9 GREEN R91 330 +3V3 GND Green — power
D10 RED R92 330 LED1 GND Red — error
D11 BLUE R93 330 LED2 GND Blue — comms
D12 GREEN R94 330 LED3 GND Green — running
D13 ORANGE R95 330 LED4 GND Orange — warning

+3V3+3V3+3V3+3V3+3V3+3V3
R127 47k SD_D3
R128 47k SD_D2
R129 47k SD_D1
R130 47k SD_D0
R131 47k SD_CMD

U25 CSNP32GCR01-AOW
DAT2 DAT3/CD CMD VDD CLK VSS DAT0 DAT1
SD_D2 SD_D3 SD_CMD SD_CLK SD_D0 SD_D1
R132 47k +3V3
SHIELD
C113 4.7u +3V3 GND
C114 100n +3V3 GND

U29 ATWINC1500-MR210UB
GPIO_6 I2C_SCL I2C_SDA RESET_N NC_5 NC_6 NC_7 NC_8 GND_1 SPI_CFG WAKE GND_2 IRQN UART_TXD SPI_RXD SPI_SSN SPI_TXD SPI_SCK UART_RXD VBAT GPIO_1 CHIP_EN VDDIO 1P3V_TP GPIO_3 GPIO_4 GPIO_5 GND_3 EP
WIFI_RST WIFI_LEN WIFI_IRQ WIFI_MOSI WIFI_CS WIFI_MISO WIFI_SCK WIFI_LEN
C106 100n +3V3 GND
C111 10u +3V3 GND
C112 10n +3V3 GND
WIFI_1P3V

J11 Conn_01x06
AUX_SPI_CS AUX_SPI_SCK AUX_SPI_MISO AUX_SPI_MOSI GND +3V3
J12 Conn_01x02

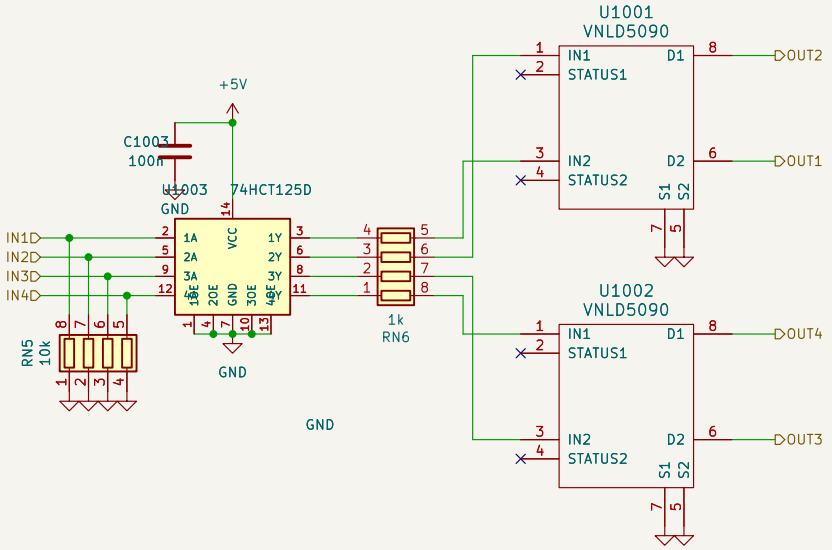
Baro pressure sensor
U24 LPS25HB
C63 100n +3V3 GND
SDA/MOSI SAO/MISO SCL/SCLK CS INT_DRDY
R87 1k +3V3
R88 1k +3V3
BARO_SDA
BARO_SCL
GND +3V3

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Sheet: /mcu/
File: mcu.kicad_sch

Title: fome−mc

Size: A3	Date: 2024−11−14	Rev: v1.0−atlas−h7
KiCad E.D.A. 10.0.7		Id: 9/21



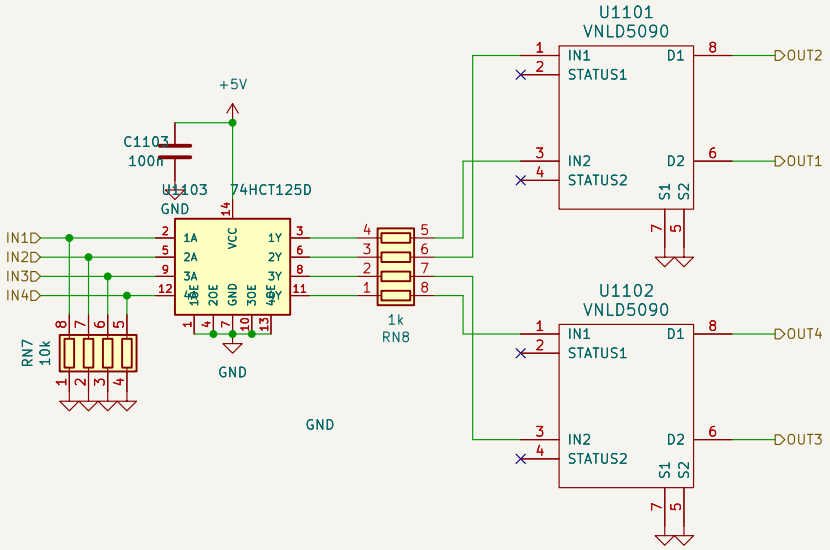
Locate output caps near connector pin

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Sheet: /lowside_quad1/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 10/21



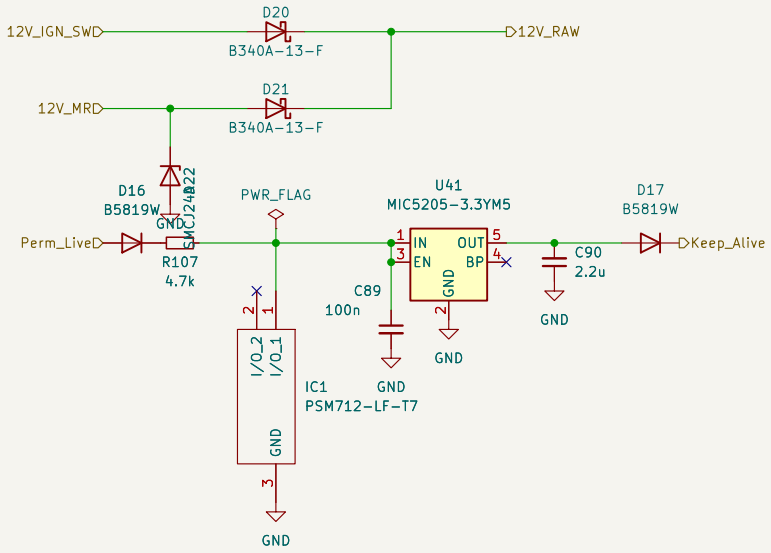
Locate output caps near connector pin

rusefi.com/s/proteus

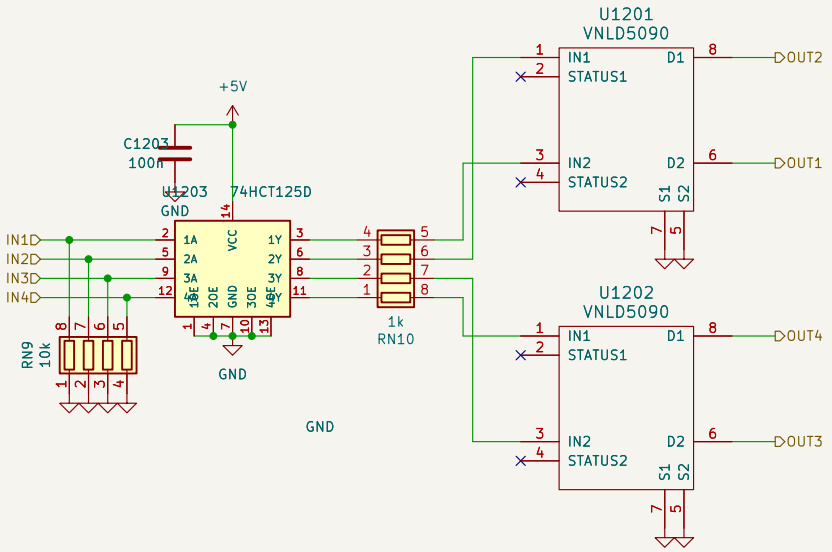
Sheet: /lowside_quad2/
File: lowside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 11/21



Sheet: /Keep_Alive/		
File: Keep_Alive.kicad_sch		
Title: fome-mc		
Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 12/21



Locate output caps near connector pin

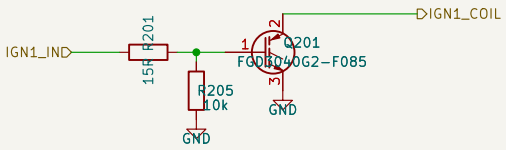
rusefi.com/s/proteus

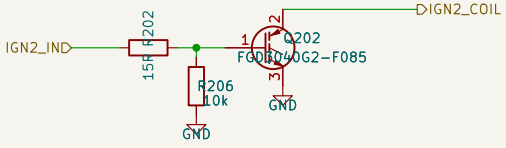
Sheet: /lowside_quad4/
File: lowside_quad.kicad_sch

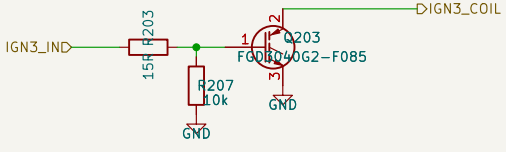
Title: fome-mc

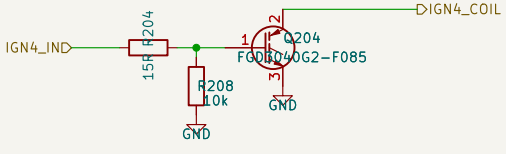
Size: A4 | Date: 2024-11-14 | **Rev: v1.0-atlas-h7**

KiCad E.D.A. 10.0.7 | Id: 13/21





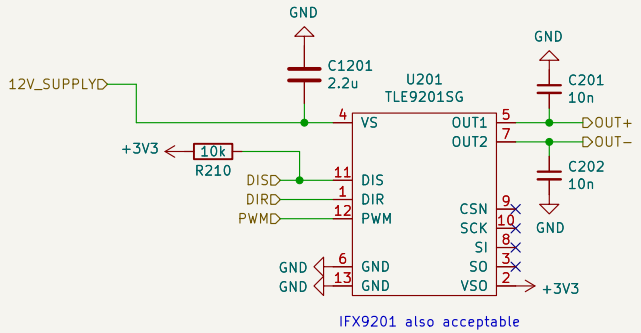




Sheet: /ign_igbt/
File: ign_igbt.kicad_sch

Title: fome-mc 4x High-Current IGBT Ignition Stage

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 14/21

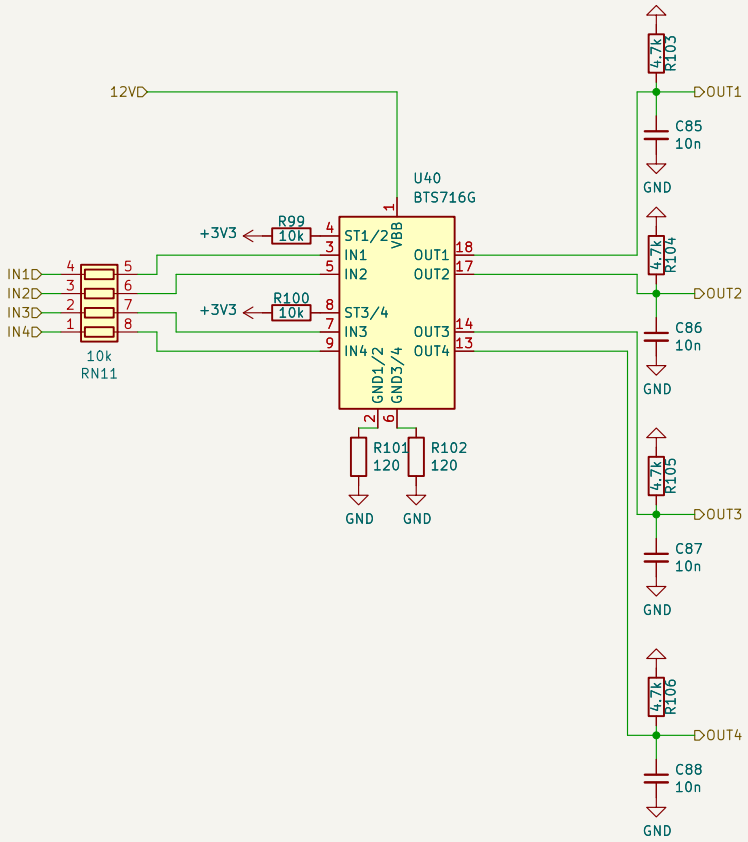


rusefi.com/s/proteus
github.com/mck1117/proteus
rusEFI

Sheet: /etb1/
File: etb-9201.kicad_sch

Title: rusEfi Proteus

Size: A4	Date: 2022-04-09	**Rev: v0.7**
KiCad E.D.A. 10.0.7		Id: 15/21

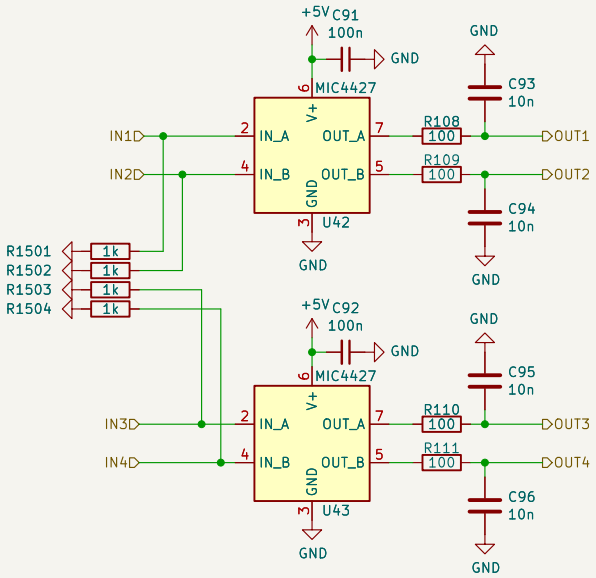


rusefi.com/s/proteus

Sheet: /highside_quad/
File: highside_quad.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 16/21

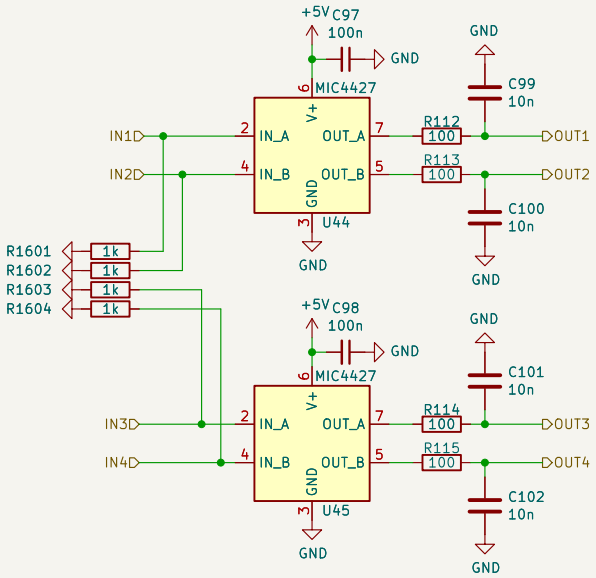


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Sheet: /ign1/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 17/21

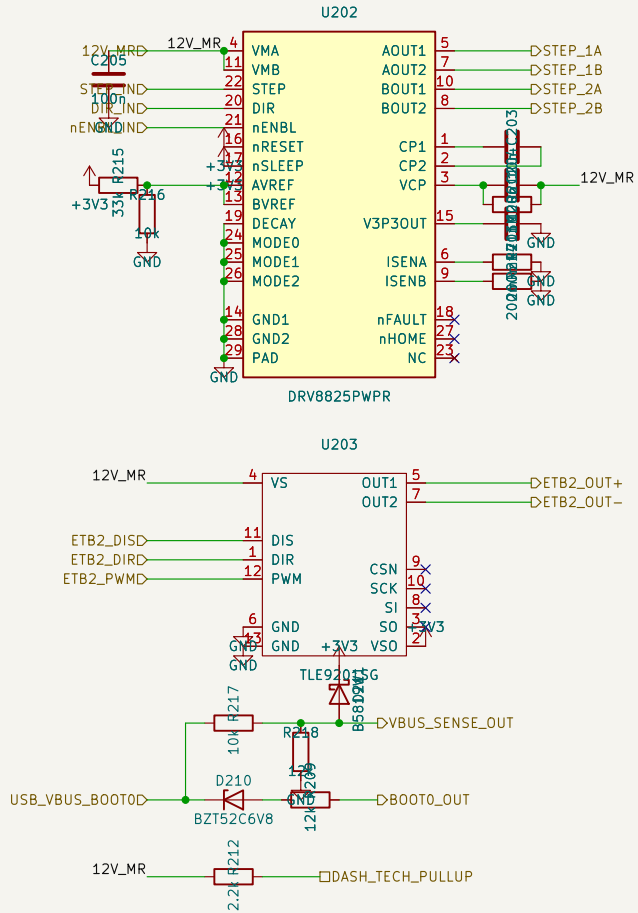


rusefi.com/s/proteus

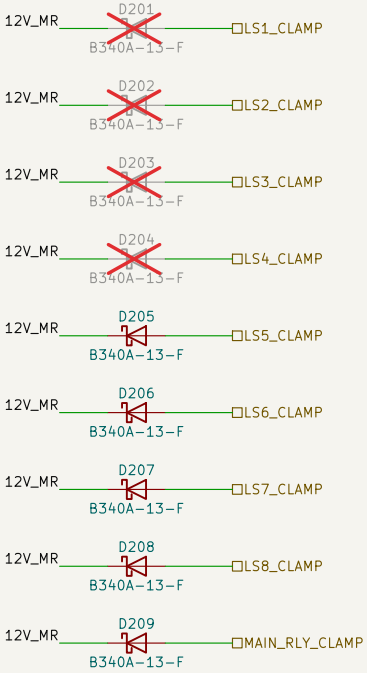
Sheet: /ign2/
File: ign4.kicad_sch

Title: fome-mc

Size: A4	Date: 2024-11-14	**Rev: v1.0-atlas-h7**
KiCad E.D.A. 10.0.7		Id: 18/21



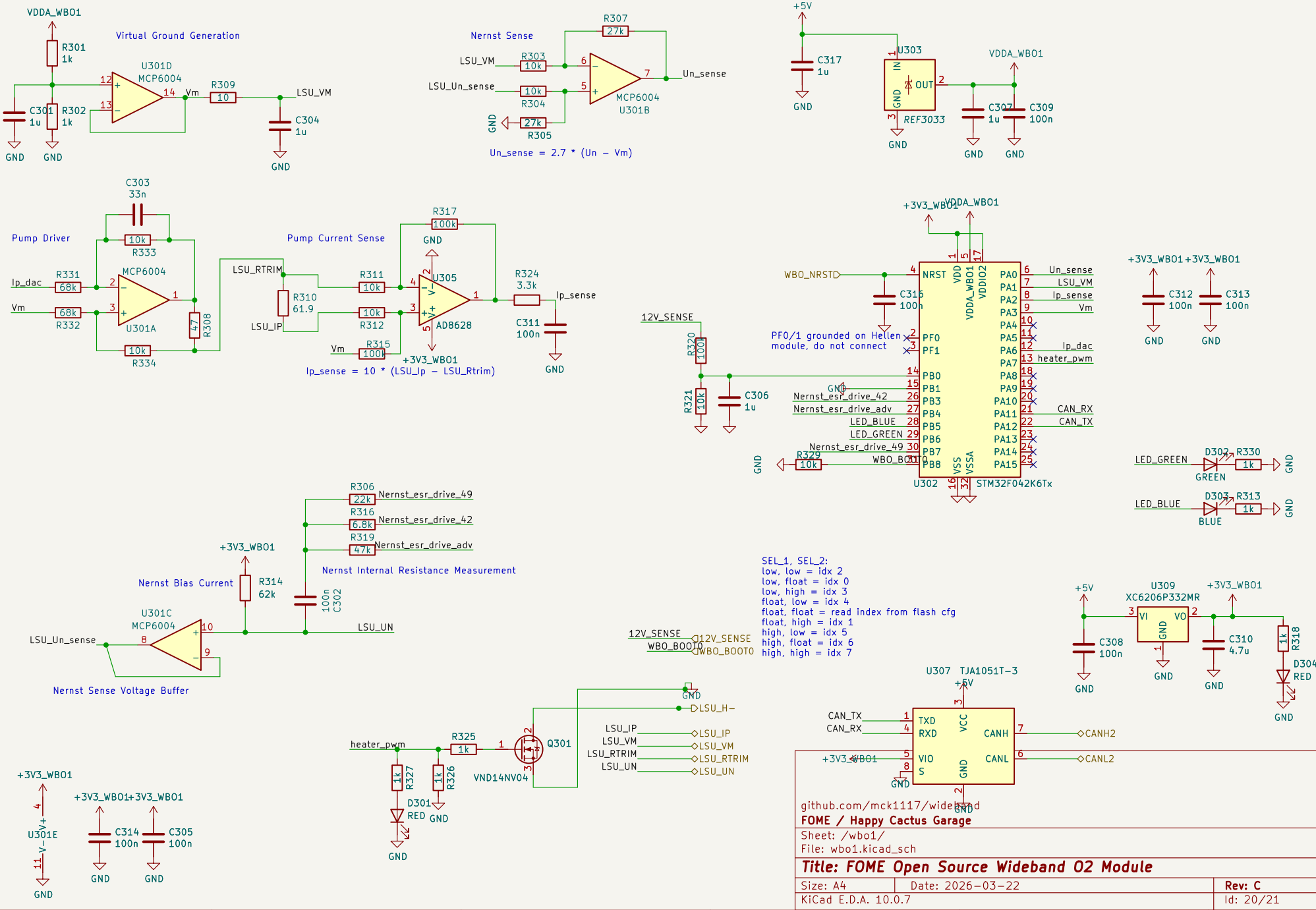
D201..D204: DNP on INJ1..4 (/LS1../LS4) so U1001/U1002 VNLD5090TR-E use internal 42V active clamp for fast injector turn-off.
D205..D209: Populated on LS5..LS8 & MAIN_RLY inductive relay/solenoid channels.

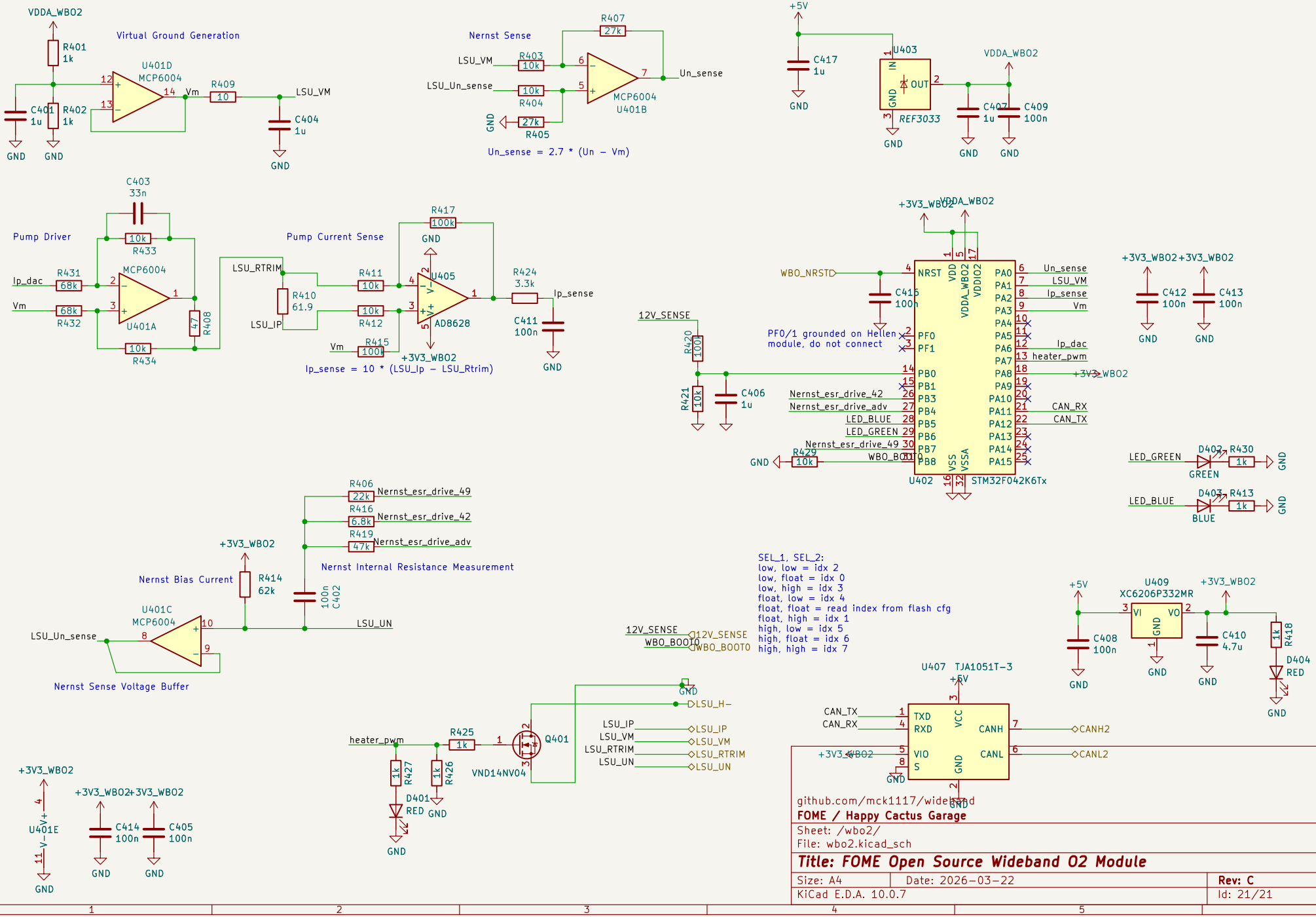


Sheet: /stepper_etb/
File: stepper_etb.kicad_sch

Title: fome-mc Stepper Driver + ETB2 H-Bridge + Low-Side Clamps + BOO

Size: A4	Date: 2026-10-16	**Rev: v1.0**
KiCad E.D.A. 10.0.7		Id: 19/21


Virtual Ground Generation
Nernst Sense
Un_sense = 2.7 * (Un − Vm)
Pump Driver
Pump Current Sense
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
Nernst Internal Resistance Measurement
Nernst Bias Current
Nernst Sense Voltage Buffer
PF0/1 grounded on Hellen module, do not connect
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
github.com/mck1117/wideband
FOME / Happy Cactus Garage
Sheet: /wbo1/
File: wbo1.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 20/21


Virtual Ground Generation
Nernst Sense
Un_sense = 2.7 * (Un − Vm)
Pump Driver
Pump Current Sense
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
PF0/1 grounded on Hellen module, do not connect
Nernst Internal Resistance Measurement
Nernst Bias Current
Nernst Sense Voltage Buffer
SEL_1, SEL_2:
low, low = idx 2
low, float = idx 0
low, high = idx 3
float, low = idx 4
float, float = read index from flash cfg
float, high = idx 1
high, low = idx 5
high, float = idx 6
high, high = idx 7
github.com/mck1117/wideband
FOME / Happy Cactus Garage
Sheet: /wbo2/
File: wbo2.kicad_sch
Title: FOME Open Source Wideband O2 Module
Size: A4
Date: 2026-03-22
Rev: C
KiCad E.D.A. 10.0.7
Id: 21/21